Test Series Code

15

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 22/04/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

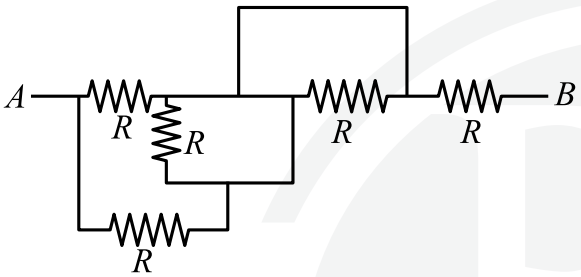
Student's Signature: _____ **Invigilator's Signature:** _____

1. A stone is projected from the origin with an initial velocity u at an angle θ above the horizontal. What is the maximum value of θ such that the straight-line distance of the stone from the starting point always increases throughout its entire motion? (Neglecting air resistance)

(1) $\sin^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (2) $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{2}{3}\right)$

2. What is the equivalent resistance between A and B in the given circuit?



(1) $\frac{3R}{2}$ (2) $2R$

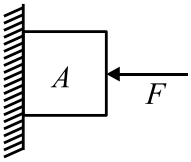
(3) R (4) $\frac{R}{2}$

3. A particle is restricted to the *xy*-plane and acted upon by a conservative force. The potential energy is given by $U(x,y)=a\left(x^2+2y^2\right)$ J, where a is a positive constant. The expression for the force vector $\vec{F}$ (in newton) is: (*x*, *y* are in metre)

(1) $-2a\left(x\hat{i}+y\hat{j}\right)$ (2) $-2a\left(x\hat{i}+2y\hat{j}\right)$

(3) $a\left(2x\hat{i}+4y\hat{j}\right)$ (4) $-a\left(x\hat{i}+2y\hat{j}\right)$

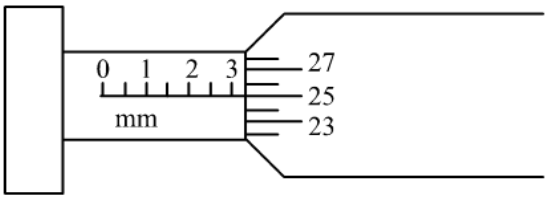
4. A block A of mass m is kept at rest against a rough vertical wall having coefficient of friction μ by applying a horizontal force F. Minimum value of F to keep the block in equilibrium is F_m, then find the acceleration of block when applied force becomes $F'=\frac{F_m}{5}$. (take g = 10 m/s²)



(1) 4 m/s² (2) 8 m/s²

(3) 10 m/s² (4) 1 m/s²

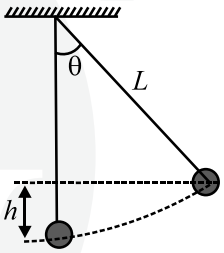
5. A screw gauge moves 1 mm on its linear scale for every 2 full rotations of its thimble, which contains 50 equal divisions. Before measurement, it was observed that when the studs are closed, the instrument has a negative zero error of 0.05 mm. If the instrument is used to measure the thickness of a metal plate as shown in the figure, the true thickness is:



(1) 3.30 mm (2) 3.20 mm

(3) 3.25 mm (4) 3.35 mm

6. A simple pendulum with bob of mass *m* and conducting wire of length *L* swings under gravity as shown in figure. An external magnetic field *B* acts in the direction perpendicular to swing. The maximum potential difference induced across the pendulum is:



(1) $2BL\sin\left(\frac{\theta}{2}\right)\sqrt{gL}$

(2) $BL\sin\left(\frac{\theta}{2}\right)\sqrt{gL}$

(3) $BL\sin\left(\frac{\theta}{2}\right)(gL)^{3/2}$

(4) $BL\sin\left(\frac{\theta}{2}\right)(gL)^{2}$

7. You are given four different parallel plate capacitors. Each is made with a different dielectric material between its plates. The table below lists the plate area (*A*), the plate separation (*d*), the dielectric constant of the material (*K*), and the dielectric strength for each capacitor.

Capacitor	Plate area (*A*)	Separation (*d*)	Dielectric Constant (*K*)	Dielectric Strength
A	2 m²	1.0 mm	3	2 MV/m
B	1 m²	0.5 mm	4	4 MV/m
C	4 m²	4.0 mm	2	1 MV/m
D	0.5 m²	0.2 mm	10	2 MV/m

Which capacitor can hold the maximum absolute amount of charge before it fails?

(1) Capacitor A (2) Capacitor B

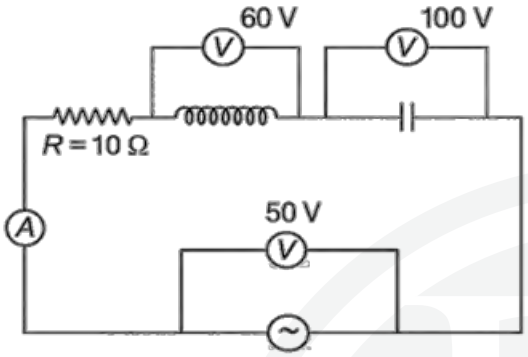
(3) Capacitor C (4) Capacitor D

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8. A mechanical heart pump circulates 6 litres of fluid per minute at a mean pressure of 100 mm of Hg . If the density of mercury is $13600\ \text{kg/m}^3$ and $g = 10\ \text{m/s}^2$, what is the approximate power output of the pump?

(1) 2.44 W (2) 0.85 W
(3) 1.36 W (4) 3.12 W

9. In the AC circuit shown, the current measured by ammeter is:



(1) 1 A (2) 2 A
(3) 3 A (4) Zero

10. A gas has a number density of 10^{14} molecules/cm^3 and a mean free path of $\frac{10^{-6}}{\sqrt{2}\pi}$ cm. The diameter of the gas molecule is:

(1) 10^{-4} cm
(2) 10^{-6} cm
(3) 10^{-3} cm
(4) 10^{-5} cm

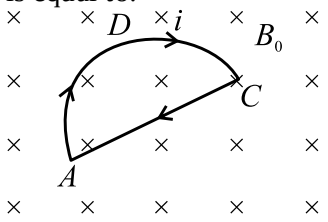
11. If a diamagnetic substance is brought near the north or the south pole of a bar magnet, it is:

(1) Repelled by the north pole and attracted by the south pole
(2) Attracted by the north pole and repelled by the south pole
(3) Attracted by both the poles
(4) Repelled by both the poles

12. A person with a normal near point (25 cm) using a compound microscope with an objective of focal length 8.0 mm and an eyepiece of focal length 2.5 cm. An object is placed at 9.0 mm from the objective If final image is formed at least distance of distinct vision, what is the magnifying power of the microscope?

(1) 88 (2) 80
(3) 44 (4) 110

13. Figure shows a conducting loop *ADCA* carrying current *i* and placed in a region of uniform magnetic field B_0. The part *ADC* forms a semicircle of radius *R*. The magnitude of force on the semicircle part of the loop is equal to:



(1) πRiB_0 (2) Zero
(3) $2\pi RiB_0$ (4) $2iRB_0$

14. In a double-slit experiment, the intensity at a point with path difference λ is I . The intensity at a point where the path difference is $\frac{\lambda}{3}$ will be:

(1) $\frac{I}{2}$ (2) $\frac{I}{4}$
(3) $\frac{I}{3}$ (4) Zero

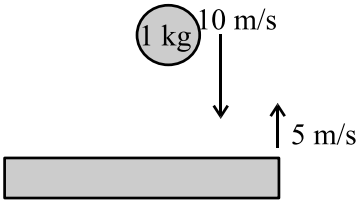
15. Four particles of equal mass *m* kept at the corners of a square and revolve in a circular orbit of radius *r* passing through these corners due to their mutual gravitational attraction. The angular velocity ω of each particle is:

(1) $\sqrt{\frac{Gm}{r^3}\left(\frac{2\sqrt{2}+1}{4}\right)}$ (2) $\sqrt{\frac{Gm}{r^3}}$
(3) $\sqrt{\frac{Gm}{r^3}\left(\frac{\sqrt{2}+1}{4}\right)}$ (4) $\sqrt{\frac{2Gm}{r^3}}$

16. Three electron energy levels of an atom are given by $E_1 = E$, $E_2 = 1.25E$, and $E_3 = 2E$. If a transition from level 3 to 1 emits a photon of wavelength λ , the wavelength of emission for the transition from level 2 to 1 is:

(1) 2λ (2) 4λ
(3) 0.25λ (4) 1.25λ

17. A ball of mass 1 kg strikes a heavy platform, elastically. The platform is moving upwards with a velocity of 5 m/s. The speed of the ball just before the collision is 10 m/s downwards. Then the magnitude of impulse imparted by the platform on the ball is:



(1) 15 N-s (2) 10 N-s
(3) 20 N-s (4) 30 N-s

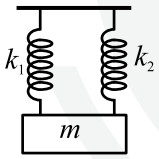
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18. Two pipes are each 2 m long. One is closed at one end (closed pipe) and the other is open at both ends (open pipe). The speed of sound in air is 340 m/s. The frequency at which both can resonate is:
(1) 340 Hz (2) 510 Hz
(3) 42.5 Hz (4) None of these

19. A plane electromagnetic wave propagates in a vacuum along the positive *z*-direction. At a specific coordinate and time, the electric field vector is measured to be 12 V/m along the positive *x*-axis. What is the magnetic field vector at that point?
(1) 4×10^{-8} T along negative *y*-direction
(2) 4×10^{-8} T along positive y-direction
(3) 3.6×10^{-9} T along positive y-direction
(4) 3.6×10^{-9} T along negative y-direction

20. An electron is moving in a circular orbit perpendicular to a uniform magnetic field of 3.57×10^{-3} T. Given the specific charge $\frac{e}{m}$ of the electron is 1.76×10^{11} C/kg, the frequency of revolution is:
(1) 10 MHz (2) 1 GHz
(3) 50 MHz (4) 100 MHz

21. A mass is suspended separately by two different springs having spring constant k_1 & k_2 and their time periods of oscillation are $t_1 = 3$ s and $t_2 = 4$ s respectively. If the same mass is connected to both springs as shown in the figure, what is the new time period t_0?



(1) 5.0 s (2) 2.4 s
(3) 7.0 s (4) 1.2 s

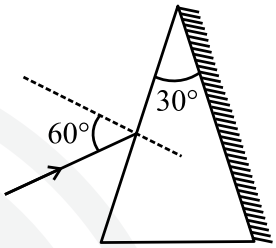
22. At what altitude from the Earth's surface does the acceleration due to gravity decrease by 75% of its value on the surface? (Let R be the radius of Earth)
(1) R (2) $2R$
(3) $\frac{R}{2}$ (4) $\frac{R}{3}$

23. A particle moves on *x*-axis according to the equation $x = x_0\sin^2\omega t$, the motion is simple harmonic:
(1) with amplitude x_0
(2) with amplitude $2x_0$
(3) with time period $(2\pi/\omega)$
(4) with time period (π/ω)

24. In a glass capillary, water rises to a height of 6 cm (angle of contact 0°). In the same tube, an unknown liquid L with a density 9 times that of water, falls by 2 cm (angle of contact 120°). The ratio of surface tension of water to surface tension of liquid L is:
(1) 1:3 (2) 3:1
(3) 1:2 (4) 1:6

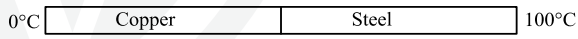
25. The volume of an air bubble quadruples as it rises from the bottom of a still pond to the surface. If the atmospheric pressure is 70 cm of Hg and the density of mercury is 10 times the density of pond water, the depth of the pond is: (Assume isothermal condition)
(1) 14 m (2) 7 m
(3) 21 m (4) 28 m

26. An isosceles prism of angle $A = 30^\circ$ has one of its surfaces silvered. Light rays falling at an angle of incidence 60° on the other surface retrace their path after reflection from the silvered surface. The refractive index of prism material is:



(1) 1.414 (2) 1.5
(3) 1.732 (4) 1.866

27. A composite bar is created by joining a Copper rod and a Steel rod in series. Both rods have identical length (L) and cross-sectional area (A). The Copper end is at 0°C and the Steel end is at 100°C. (Given: $K_{Cu} = 400$ W m^{-1} K^{-1} and $K_{St} = 100$ W m^{-1} K^{-1})



Match the physical quantities in **List-I** with their numerical values in **List-II**:

List-I		List-II	
(A)	Equivalent thermal conductivity (W m^{-1} K^{-1})	(I)	0.25
(B)	Steady-state junction temperature (°C)	(II)	800
(C)	Ratio of thermal resistances ($R_{Cu} : R_{St}$)	(III)	160
(D)	Total heat flow rate in Watts (given $\frac{A}{L} = 0.1$ m)	(IV)	20

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-I, D-III

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28. C and Si both have same lattice structure, having 4 bonding electrons in each. However, C is insulator whereas Si is intrinsic semiconductor. This is because:

(1) The four bonding electrons in the case of C lie in the second orbit, whereas in the case of Si they lie in the third, therefore forbidden energy gap in C is very large while for Si it is relatively small.

(2) The four bonding electrons in the case of C lie in the third orbit, whereas for Si they lie in the fourth orbit, therefore forbidden energy gap in C is small while for Si it is relatively large.

(3) In case of C the valence band is not completely filled at absolute zero temperature.

(4) In case of C the conduction band is partially filled even at absolute zero temperature.

29. Two balls of same material and identical emissivity have their diameters in the ratio 1 : 2. They are heated to the same temperature and are left in a room to cool by radiation, then the initial rate of loss of heat:

(1) Will be same for the balls

(2) For larger ball is half that of other ball

(3) For larger ball is twice that of other ball

(4) For larger ball is four times that of the other ball

30. The following question, a statement of Assertion is followed by a Reason. Select the correct alternative.

Assertion (A): Every metal has a definite work function, still all photoelectrons do not come out with the same energy if incident radiation is monochromatic.

Reason (R): Work function is the minimum energy required for the electron in the highest level of the conduction band to get out of the metal. Not all electrons in the metal belong to this level rather they occupy a continuous band of levels.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

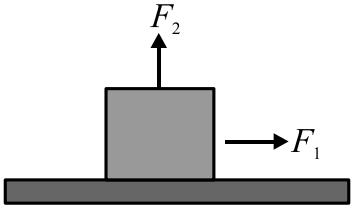
(4) Both A and R are true and R is NOT the correct explanation of A.

31. A straight structural rod of circular cross-section has a radius $r = 10$ mm and a length $l = (2\pi)$ m. When a tensile force of $F = 40$ kN is applied along its length, determine the elastic potential energy stored in the rod. (Young's modulus $Y = 2.0 \times 10^{11}$ N m^{-2})

(1) 160 J (2) 40 J

(3) 80 J (4) 20 J

32. In the figure shown, horizontal force F_1 is applied on a block but the block does not slide. Then as the magnitude of vertical force F_2 is increased from zero until the block begins to slide, consider the following statements:



I. The magnitude of normal reaction on the block increases.

II. Before sliding, static frictional force acting on the block remains constant.

III. Value of limiting friction decreases.

Which of the statements is/are **correct**?

(1) I and II only (2) II and III only

(3) III only (4) I and III only

33. A particle of mass m is thrown with an initial speed u at an angle θ with the horizontal. Find the ratio of the magnitude of the torque of its weight to the magnitude of its angular momentum, both calculated about the point of projection when the particle is at its highest point. (g is acceleration due to gravity)

(1) $\frac{g\cos\theta}{u\sin^2\theta}$ (2) $\frac{g}{2u\sin\theta}$

(3) $\frac{g}{u\cos\theta}$ (4) $\frac{2g}{u\sin\theta}$

34. The binding energies of the nuclei A and B are E_a and E_b respectively. Three nuclei atoms of the element B fuse to give single nucleus of element A and an energy Q is released. Then E_a, E_b and Q are related as:

(1) $E_a - 3E_b = Q$

(2) $3E_b - E_a = Q$

(3) $E_a + 3E_b = Q$

(4) $E_b + 3E_a = Q$

35. The displacement y of an object as a function of time t is expressed by the equation: $y = a\sin(\pi t) + bt + ct^2\cos\left(\frac{\pi}{2}t\right)$ where y is in meters and t is in seconds. The dimensions of the ratio $\frac{ac}{b}$ are identical to those of:

(1) $\frac{y}{t}$ (2) y

(3) $\frac{y}{t^2}$ (4) yt

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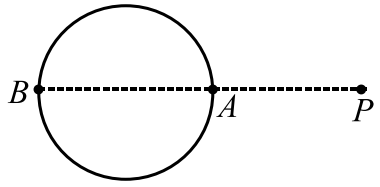
36. A Carnot engine absorbs 12 kJ of heat from a high-temperature reservoir at 227°C and rejects some heat to a sink at 127°C. The work output of the engine per cycle is:
(1) 3.6 kJ (2) 2.4 kJ
(3) 1.2 kJ (4) 4.8 kJ

37. Consider the following logic gate configurations and evaluate their outputs:
I. An AND gate with inputs 0 and 1.
II. A NOR gate with inputs 0 and 1.
III. A NAND gate with inputs 1 and 1.
IV. An OR gate with inputs 0 and 1.
Which of the following statements is **correct** regarding the outputs of these gates?
(1) Gates I and III have an output of 1, while II and IV have an output of 0.
(2) Gate IV has an output of 1, while gates I, II, and III have an output of 0.
(3) Gates II and IV have an output of 1, while I and III have an output of 0.
(4) All gates have an output of 1.

38. Two runners start simultaneously from the same point on a circular track of radius 50 cm. They run in the same direction with speeds of 25π cm/s and 20π cm/s. After how many seconds will they meet for the first time?
(1) 40 s (2) 10 s
(3) 50 s (4) 20 s

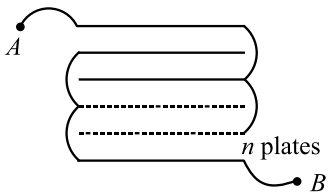
39. Two bodies of mass 1 kg and 2 kg have position vectors $\left(2\hat{i}+5\hat{j}+3\hat{k}\right)$ m and $\left(2\hat{i}-\hat{j}\right)$ m respectively. The magnitude of the position vector of the center of mass of this system is equal to the magnitude of which of the following vectors?
(1) $\left(2\hat{i}+3\hat{j}+\hat{k}\right)$m (2) $\left(\hat{i}+2\hat{j}+\hat{k}\right)$m
(3) $\left(2\hat{i}-\hat{k}\right)$m (4) $\left(3\hat{i}-\hat{j}+\hat{k}\right)$m

40. A hollow conducting sphere is placed in an electric field produced by a point charge at P as shown in the figure. Let V_A and V_B be the net electrostatic potential at points A and B respectively. Then,



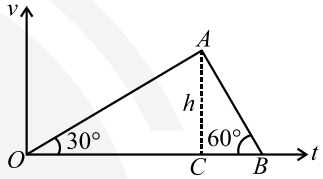
(1) $V_A > V_B$
(2) $V_B > V_A$
(3) $V_B = V_A$
(4) Relation cannot be determined with the given information

41. In the arrangement of n identical parallel plates shown, each of area A and separated by a distance d from the next, alternate plates are joined together to terminals A and B respectively, find the equivalent capacitance C_{AB}.



(1) $\frac{(n-1)\epsilon_0 A}{d}$ (2) $\frac{(n-2)\epsilon_0 A}{d}$
(3) $\frac{\epsilon_0 A}{(n-1)d}$ (4) $\frac{\epsilon_0 A}{nd}$

42. The velocity-time $(v-t)$ graph of a body moving along straight line is shown in the figure. For the time intervals OC and CB, the ratio of the distances covered is:



(1) 3:1 (2) 1:3
(3) $\sqrt{3}:1$ (4) $\sqrt{3}:2$

43. In a certain region, a radial electric field is described by the function $\vec{E}=(500r)\hat{r}$ V/m, where r is the distance from the origin. Calculate the total charge enclosed within a spherical surface of radius 0.2 m centered at the origin.
(1) $(8\pi\epsilon_0)$ C (2) $(16\pi\epsilon_0)$ C
(3) $(4\pi\epsilon_0)$ C (4) $(32\pi\epsilon_0)$ C

44. Two stationary point charges $+Q$ and $+4Q$ are separated by a distance L in vacuum. At what distance from the smaller charge $(+Q)$ on the line joining them is the net electric field zero?
(1) $\frac{L}{2}$ (2) $\frac{L}{5}$
(3) $\frac{L}{3}$ (4) $\frac{2L}{3}$

45. Two incandescent lamps labeled 120V,40W and 120V,100W are connected in series across a 120V power line. Which statement regarding the potential drop across the lamps is **correct**?
(1) Potential drop is identical for both lamps.
(2) The 40W lamp has a larger potential drop than the 100W lamp.
(3) The 100W lamp has a larger potential drop than the 40W lamp.
(4) The total potential drop across the pair is 240V.

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46. In the reaction of benzaldehyde and acetone to prepare 585g of dibenzalacetone, 145 g of acetone is used. The amount of benzaldehyde required in moles is:
(1) 1.5 mol (2) 5.0 mol
(3) 3.0 mol (4) 4.5 mol

47. The number of orbitals in h-type subshell is:
(1) 9 (2) 11
(3) 13 (4) 15

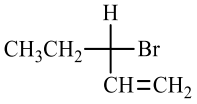
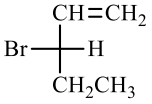
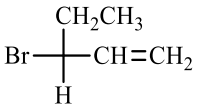
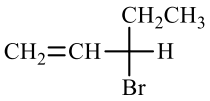
48. Consider the following statements:
(A) Electron density in the XY plane for $3d_{xy}$ orbital is non zero.
(B) Electron density in the XY plane for $3d_{z^2}$ orbital is zero.
(C) 3s orbital has two spherical nodes.
(D) For $2p_z$ orbital, XY is the nodal plane.
Which of these are **incorrect** statements?
(1) A and B (2) B and C
(3) Only B (4) A and C

49. Which reaction, with the following values of ΔH and ΔS at 300 K, is spontaneous and endothermic?
(1) $\Delta H = -36$ kJ, $\Delta S = +120$ J K^{-1}
(2) $\Delta H = -36$ kJ, $\Delta S = -120$ J K^{-1}
(3) $\Delta H = +36$ kJ, $\Delta S = +160$ J K^{-1}
(4) $\Delta H = +36$ kJ, $\Delta S = -160$ J K^{-1}

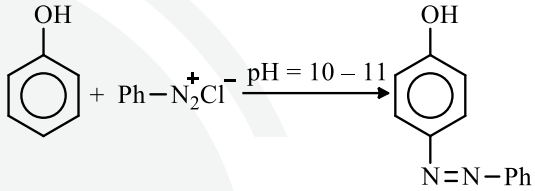
50. What is the threshold wavelength of the metal if light of wavelength 4000 Å ejects photoelectrons with speed 6×10^5 m s^{-1}?
(1) 5946 Å (2) 4000 Å
(3) 3100 Å (4) 7200 Å

51. The work done by a system is 12 J when 30 J heat is supplied to it. The change in internal energy is:
(1) 18 J
(2) 42 J
(3) –18 J
(4) 12 J

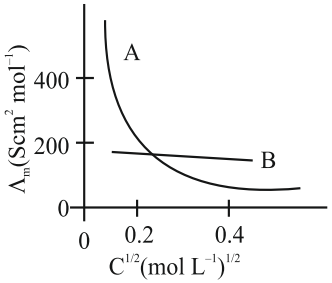
52. For ammonium phosphate solution, if pK_a of phosphoric acid is 5.23 and pK_b of ammonium hydroxide is 4.75, the nature of the solution is:
(1) Strongly acidic (pH = 3)
(2) Neutral (pH = 7)
(3) Slightly basic (pH = 7.24)
(4) Slightly basic (pH = 7.48)

53. Which of the following compounds has R configuration?
(1) 
(2) 
(3) 
(4) 

54. Which of the following gives a black precipitate when H_2S is passed through an acidic solution?
(1) Zinc acetate
(2) Copper acetate
(3) Magnesium acetate
(4) Calcium acetate

55. Which is **correct** about reaction?

(1) C–N coupling reaction: Carbanion is intermediate.
(2) N–N coupling reaction: Carbocation is intermediate.
(3) C–N coupling reaction: Carbocation is intermediate.
(4) N–N coupling reaction: Carbanion is intermediate.

56. A 5% (w/w) glucose solution has density 1.20 g mL^{-1}. Its strength in g L^{-1} is:
(1) 50 (2) 60
(3) 75 (4) 90

57. The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are:


	A	B
(1)	NH_4OH	NH_4OH
(2)	KCl	KCl
(3)	NH_4OH	KCl
(4)	KCl	NH_4OH

[Contd...

58. For the reaction $H_2F_2(g) \rightarrow H_2(g) + F_2(g)$ if $\Delta U = -59.6\ kJ\ mol^{-1}$ at 27°C, then the value of $\Delta H - \Delta U$ is:
(1) 0.83 kJ mol^{-1} (2) 1.66 kJ mol^{-1}
(3) 2.49 kJ mol^{-1} (4) 4.98 kJ mol^{-1}

59. Half-life does **not** vary with initial concentration for:
(1) Zero order reaction
(2) First order reaction
(3) Second order reaction
(4) Third order reaction

60. The number of unpaired electrons in $[Fe(CN)_6]^{4-}$ and $[FeF_6]^{4-}$ respectively is:
(1) 0 and 0 (2) 2 and 4
(3) 0 and 4 (4) 4 and 4

61. The products formed in the following reaction:
$C_6H_5-O-CH_3 + HI \xrightarrow{heat}$ are
(1) C_6H_5 – I and CH_3–OH
(2) C_6H_5–OH and CH_3–I
(3) C_6H_5–CH_3 and HOI
(4) C_6H_6 and CH_3OI

62. For the mixture of non-reacting gases X (0.6 g) and Y (0.45 g) having total pressure 740 mm Hg, what is the partial pressure of gas Y?
(Given: molar mass of X = 20 g mol^{-1} and Y = 45 g mol^{-1})
(1) 185 mm Hg (2) 555 mm Hg
(3) 370 mm Hg (4) 740 mm Hg

63. Which of the following elements is **not** an actinoid?
(1) Neptunium (2) Curium
(3) Thorium (4) Europium

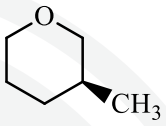
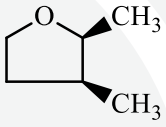
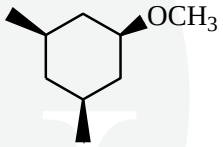
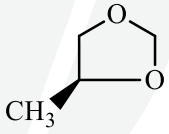
64. An ideal solution is one which:
(1) shows positive deviation from Raoult's law
(2) shows negative deviation from Raoult's law
(3) obeys Raoult's law over the entire range
(4) has zero vapour pressure

65. During kinetic study of reaction 2A + B → C + D, the following result were obtained.

	A[M]	B[M]	Initial Rate of Formation of D
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Base on above data, order of the reaction is with respect to B is:
(1) 3 (2) 1
(3) 2 (4) 4

66. The reaction,
$\frac{1}{2}H_2(g) + AgBr(s) \rightarrow H^+(aq) + Br^-(aq) + Ag(s)$,
occurs in the galvanic cell:
(1) $Pt\,|\,H_2(g)\,|\,HBr(aq)\,||\,AgNO_3(aq)\,|\,Ag$
(2) $Pt\,|\,H_2(g)\,|\,HBr(aq)\,|\,AgBr(s)\,|\,Ag$
(3) $Ag\,|\,AgBr(s)\,|\,KBr(aq)\,||\,AgNO_3(aq)\,|\,Ag$
(4) $Pt\,|\,H_2(g)\,|\,KBr(aq)\,|\,AgBr(s)\,|\,Ag$

67. Which compound have two chiral centre?
(1) 
(2) 
(3) 
(4) 

68. Number of moles of bromine required to completely react with 5.0 g of pent-1-ene is:
(1) 0.0714 mol (2) 0.143 mol
(3) 0.0500 mol (4) 0.0286 mol

69. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In the photoelectric effect, electrons are emitted from a metal surface immediately when light of frequency greater than the threshold frequency falls on it.
Reason R: A single photon can transfer its energy to a single electron during interaction.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both A and R are correct but R is not the correct explanation of A.
(2) A is correct but R is not correct.
(3) Both A and R are correct and R is the correct explanation of A.
(4) A is not correct but R is correct.

[Contd...

70. Among the following, the total number of species **not** having eight electrons around the central atom in its outermost shell is:

NH_3, BCl_3, $BeCl_2$, CO_2, PF_5

(1) 2
(2) 3
(3) 4
(4) 5

71. An organic compound contains 80% by mass carbon and the remaining hydrogen. The **correct** empirical formula is:

(1) CH_4
(2) CH_3
(3) CH_2
(4) CH

72. In a multielectron atom, which of the following orbitals described by three quantum numbers will have the same energy in the absence of electric and magnetic fields?

A. $n = 2, \ell = 1, m_\ell = 0$
B. $n = 2, \ell = 1, m_\ell = -1$
C. $n = 3, \ell = 1, m_\ell = 0$
D. $n = 3, \ell = 2, m_\ell = 1$

Choose the **correct** answer from the options given below:

(1) A and B only
(2) B and C only
(3) C and D only
(4) A and D only

73. The equation

$$\log k = \log 10^{-12} - \frac{26000}{T}$$

is obeyed for decomposition of compound A. The activation energy E_a(in kJ) is:

(1) 498 kJ
(2) 49 kJ
(3) 216 kJ
(4) 598 kJ

74. The resistance of a conductivity cell containing 0.01 M KCl solution at 298 K is 1750 Ω. If the conductivity of the solution is 0.152×10^{-3} S cm^{-1}, the conductance and cell constant of the cell is:

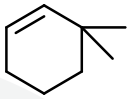
(1) 5.71×10^{-4} S and 2.66×10^{-1} cm^{-1}
(2) 1.75×10^{3} S and 5.71×10^{-4} cm^{-1}
(3) 2.66×10^{-1} S and 1.75×10^{3} cm^{-1}
(4) 8.68×10^{-8} S and 5.71×10^{-4} cm^{-1}

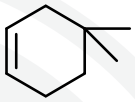
75. For an oxidation-reduction reaction in which 3 electrons are transferred, if $\Delta G° = 17.37$ kJ mol^{-1} at 25°C, the value of n E^{o}_{cell} is:

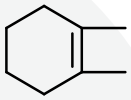
(1) 0.18
(2) –0.18
(3) 0.06
(4) –0.06

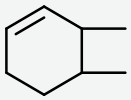
76. 

Major product is:

(1) 

(2) 

(3) 

(4) 

77. The difference between electron gain enthalpies will be maximum between:

(1) Ne and F
(2) Ne and Cl
(3) F and Cl
(4) Ar and F

78. How many dichloro derivatives (excluding stereoisomerism) can be obtained from cyclobutane?

(1) 3 (2) 4
(3) 5 (4) 6

79. Number of elements from the following that **cannot** show valency more than their group valency in compounds is:

B, C, N, O, F, P, S, Cl, Si

(1) 3 (2) 4
(3) 5 (4) 6

80. The hybridisation of nitrogen in  is;

(1) sp (2) sp^2
(3) sp^3 (4) sp^3d

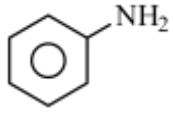
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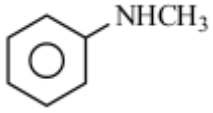
81. The **correct** increasing order of the oxidation states of sulphur in S_8, S_2F_2 and H_2S is:

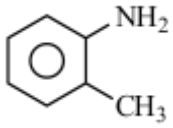
(1) $S_2F_2 < H_2S < S_8$ (2) $S_2F_2 < S_8 < H_2S$

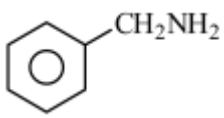
(3) $H_2S < S_8 < S_2F_2$ (4) $H_2S < S_2F_2 < S_8$

82. Phthalimide on treatment with ethanolic KOH forms X which on heating with benzyl chloride followed by alkaline hydrolysis gives:

(1) 

(2) 

(3) 

(4) 

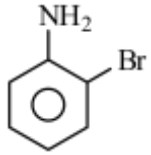
83. If 3.69 g of glucose is dissolved in 100 g of water, the lowering in vapour pressure of water at room temperature will be:

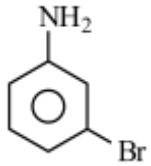
(Assume dilute solution is formed; vapour pressure of pure water = 54.2 mm Hg; molar mass of glucose 180 g mol^{-1})

(1) 0.10 mm Hg (2) 0.20 mm Hg

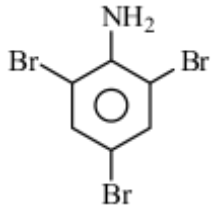
(3) 0.40 mm Hg (4) 0.54 mmHg

84. Aniline reacts with bromine water at room temperature to give a white precipitate of:

(1) 

(2) 

(3) 

(4) 

85. Paper chromatography is an example of

(1) Partition chromatography

(2) Thin layer chromatography

(3) Column chromatography

(4) Adsorption chromatography

86. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Dumas method	(I)	Silica gel
(B)	Kjeldahl's method	(II)	$AgNO_3$
(C)	Chromatography	(III)	Ammonium sulphate
(D)	Carius method	(IV)	Nitrogen gas

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-II, C-IV, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-II, C-III, D-IV

87. The balanced chemical reaction among the following is:

(1) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

(2) $2MnO_4^- + 5C_2O_4^{2-} + 8H^+ \rightarrow Mn^{2+} + 10CO_2 + 4H_2O$

(3) $2MnO_4^- + C_2O_4^{2-} + 2H^+ \rightarrow 2Mn^{2+} + 2CO_2 + H_2O$

(4) $MnO_4^- + C_2O_4^{2-} + 8H^+ \rightarrow Mn^{2+} + 2CO_2 + 8H_2O$

88. The **correct** formula of potash alum is;

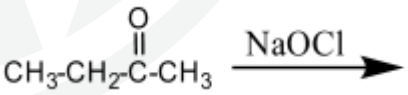
(1) $K_2SO_4 \cdot (NH_4)_2SO_2 \cdot 12H_2O$

(2) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(3) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

(4) $Na_2SO_4 \cdot Al_2(SO_4)_3 \cdot 6H_2O$

89. What is the product formed in the given reaction?



(1) $CHCl_3$ (2) CH_3Cl

(3) CH_3CH_2Cl (4) CH_2Cl_2

90. Given below are two statements:

Statement I: Alcohols and phenols can be distinguished by neutral ferric chloride.

Statement II: Phenols give violet colour complex with neutral $FeCl_3$.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

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91. A linear DNA fragment contains three recognition sites for enzyme X and two recognition sites for enzyme Y. If the DNA is fully digested with both enzymes simultaneously, what is the maximum number of fragments that can be observed on an agarose gel, and what determines the visibility of these fragments?

(1) 5 fragments; determined by the sequence of the DNA.
(2) 6 fragments; determined by the circularity of the original DNA.
(3) 6 fragments; determined by the molecular weight of the fragments.
(4) 12 fragments; determined by the voltage applied during electrophoresis.

92. A newly discovered organism has haplodiploid mechanism of sex determination. Which of the following cannot be true about it?

(1) Three pairs of jointed legs present
(2) Production of gametes in males by mitosis
(3) Females have double the number of chromosomes as compared to males
(4) Placed in phylum Chordata

93. The decomposers in an aquatic ecosystem are;

(1) marginal plants.
(2) bacteria and fungi.
(3) primary consumers.
(4) present at the third trophic level.

94. Match **List-I** with **List-II**.

List-I (Structure)		**List-II (Function in Bacteria)**	
(A)	Plasmid	(I)	Cell wall formation
(B)	Mesosome	(II)	Protection (Part of cell envelope)
(C)	Glycocalyx	(III)	Antibiotic resistance
(D)	Fimbriae	(IV)	Attachment to substratum

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

95. Match **List-I** with **List-II.**

List-I (Animal)		**List-II (Excretory waste)**	
(A)	*Labeo*	(I)	Urea
(B)	*Carcharadon*	(II)	Ammonia
(C)	*Rana*	(III)	Uric acid
(D)	*Calotes*	(IV)	Ammonia in larva and urea in adult

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

96. Consider the following statements:

Statement I: In blue-white screening, recombinant colonies produce blue color on a chromogenic substrate.

Statement II: Transformants are cells that have acquired a foreign DNA (often plasmid DNA).

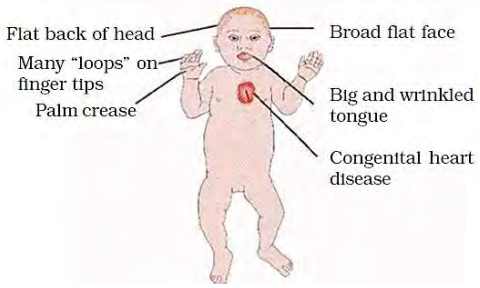
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

97. In *Oxalis*;

(1) roots come out of the ground and grow vertically upwards.
(2) leaf is modified into a pitcher.
(3) stem is modified for spread and vegetative propagation.
(4) leaves have lost chlorophyll in the course of evolution.

98. An individual inflicted with a particular genetic disorder is shown below. Choose the correct statement about it.



(1) It is a case of aneuploidy.
(2) The number of autosomes in this individual is 44.
(3) The number of sex chromosomes in this individual is 3.
(4) This occurs only in females.

[Contd...

99. In *Cedrus*,
(1) the female gametophyte is called archegonia.
(2) true fruits are produced.
(3) the stems are branched.
(4) the male gametophyte can exist independently.

100. The interfascicular cambium in plants;
(1) appears in early stages of life.
(2) is responsible for root and stem elongation in length.
(3) help in regenerating the parts of grass removed by the grazing herbivores.
(4) is a type of cylindrical meristem.

101. An experimental stirred-tank bioreactor is superior to shake flasks for large-scale enzyme production because it can:
(1) Maintain cells in the lag phase for longer durations.
(2) Avoid the need for any downstream processing.
(3) Maintain cells in the exponential phase with optimal growth conditions.
(4) Grow genetically unstable cells without selective pressure.

102. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Urochordata	(I)	Notochord persists throughout life from head to tail
(B)	Cephalochordata	(II)	Notochord present only in larval tail
(C)	Cyclostomata	(III)	Sucking and circular mouth without jaws
(D)	Osteichthyes	(IV)	Skin is covered with cycloid/ ctenoid scales

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

103. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Tryptophan	(I)	Drug
(B)	Aspartic acid	(II)	Toxin
(C)	Curcumin	(III)	Aromatic amino acid
(D)	Ricin	(IV)	Acidic amino acid

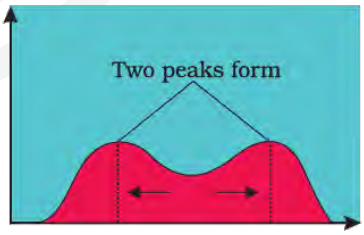
Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-IV, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

104. Oval or round shaped cells are present in;
(1) the cortical layer that has casparian strips in dicot roots.
(2) the layer present on the inner side of endodermis but above phloem in dicot stem.
(3) abaxially placed mesophyll of dorsiventral leaves.
(4) the structure that regulate the process of transpiration and gaseous exchange in monocots.

105. Which of the following is an **incorrect** match for the common Indian frog (*Rana tigrina*)?
(1) Cutaneous respiration – Exchange of gases through skin in water
(2) Pulmonary respiration – Respiration by lungs on land
(3) Fertilisation – Internal and occurs in the cloaca
(4) Aestivation – Summer sleep in deep burrows

106. A woman with normal menstrual cycles has a complete blockage in the ampullary region of both oviducts. What is the most likely consequence?
(1) Fertilisation can still occur in the isthmus.
(2) Menstruation will cease immediately.
(3) Fertilisation is unlikely because the normal site of fusion is obstructed.
(4) Ovulation will stop due to the blockage.

107. Which of the following is an example of type of natural selection shown by below diagram?



(1) Industrial melanism in the peppered moth (*Biston betularia*) where dark moths survived better in polluted areas.
(2) Human birth weight, where very low and very high birth weights have higher mortality rates compared to average weights.
(3) A population of finches where birds with small beaks and birds with large beaks are favored over those with intermediate-sized beaks.
(4) Long-necked giraffes being favored over short-necked giraffes as they can reach higher foliage.

108. Given below are two statements:
Statement I: The process of sperm transformation into spermatozoa is called spermiation.
Statement II: In females, primary oocytes are arrested at Prophase I of meiosis until puberty.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

109. Degeneration of corpus luteum triggers which of the following event during menstrual cycle?
(1) Ovulation (2) Proliferative phase
(3) Luteal phase (4) Menstrual bleeding

110. Which statement regarding the cockroach's digestive system is **incorrect**?
(1) The entire foregut is lined by cuticle.
(2) The gizzard has thick inner cuticle forming six highly chitinous teeth.
(3) Hepatic caeca are present at the junction of foregut and midgut.
(4) Crop is used for digestion of food.

111. Which of the following is the **correct** ordered sequence of fossil transitions across water-land interfaces?
(1) Amphibians → lobefin fish → reptiles → mammals
(2) Jawless fish →amphibians → lobefin fish → reptiles
(3) Reptiles → amphibians → lobefin fish → birds
(4) Jawless fish → lobefin fish → amphibians → reptiles

112. Morphine is extracted from the latex of which plant?
(1) *Cannabis sativa*
(2) *Atropa belladonna*
(3) *Papaver somniferum*
(4) *Erythroxylum coca*

113. The presence of a pathogen in a patient's body is normally suspected only when the pathogen has produced a disease symptom. However, very low concentrations can be detected early using:
(1) Physical examination
(2) Gram staining
(3) Serum analysis
(4) PCR (Polymerase Chain Reaction)

114. Match the following animal symmetries and coelom types with their respective Phyla in the options given below.
- Feature A: Asymmetrical, Acoelomate
- Feature B: Radial symmetry, Acoelomate
- Feature C: Bilateral symmetry, Pseudocoelomate
- Feature D: Bilateral symmetry, Coelomate

(1) A-Porifera, B-Echinodermata (larva), C-Aschelminthes, D- Chordata
(2) A-Ctenophora, B-Platyhelminthes, C-Annelida, D-Aschelminthes
(3) A-Porifera, B-Ctenophora, C-Aschelminthes, D-Annelida
(4) A-Coelenterata, B-Ctenophora, C-Platyhelminthes, D-Arthropoda

115. Who amongst the following scientists was not involved in developing the entire body of molecular biology after Mendel's experiments?
(1) Brenner
(2) Benzer
(3) Morgan
(4) Nirenberg

116. Which of the following is seen in most monocots?
(1) Activity of cork cambium
(2) Bulliform cells on the abaxial epidermis of leaves
(3) Large and well developed pith in roots
(4) Development of phloem parenchyma in stem at later stage of life

117. The human "need turned into greed" led to the extinction of;
(1) *Lantana*.
(2) Steller's sea cow.
(3) *Eicchornia*.
(4) Dodo.

118. Regarding the human skeletal system, which of the following is **correct**?
(1) The axial skeleton comprises 126 bones.
(2) Cranial bones are 14 in number.
(3) The first seven pairs of ribs are true ribs attached to the sternum.
(4) Limbs have 206 bones in total.

119. Match **List-I** with **List-II**.

List-I (Trait)		List-II (Characteristic inheritance)	
(A)	Seed colour in pea	(I)	Polygenic inheritance
(B)	Flower colour in snapdragon	(II)	Dominance
(C)	Skin colour in humans	(III)	X-linked
(D)	Color blindness in humans	(IV)	Incomplete dominance

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-IV, B-I, C-II, D-III

120. The smaller sub unit of prokaryotic ribosome is;
(1) 80S.
(2) 60S.
(3) 40S.
(4) 30S.

121. Identify the **correct** statement regarding cancer biology.
(1) Benign tumors show metastasis and rapidly invade surrounding tissues.
(2) Malignant tumors consist of neoplastic cells that can form secondary tumors at distant sites.
(3) Contact inhibition is a characteristic feature of cancer cells.
(4) Proto-oncogenes are viral genes that originate from oncogenic viruses.

122. The population interaction where one species is benefitted and the other is neither benefitted nor harmed is called:
(1) Mutualism.
(2) Predation.
(3) Amensalism.
(4) Commensalism.

123. Match **List-I** with **List-II**.

List-I (Scientist)		List-II (Experimental material)	
(A)	T. H Morgan	(I)	*Streptococcus pneumoniae*
(B)	Frederick Griffith	(II)	*E. coli*
(C)	Jacob and Monod	(III)	*Vicia faba*
(D)	Taylor and colleagues	(IV)	Fruitfly

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-II, C-III, D-IV

124. Read the following regarding PCR:
A. It requires a thermostable DNA polymerase (*Taq* polymerase).
B. Primers are short, double-stranded DNA fragments.
C. It can amplify a specific DNA segment to about a billion copies in 30 cycles.
D. Denaturation involves heating the DNA to join strands.
E. Annealing involves cooling to allow primers to bind.
Choose the **correct** ones:
(1) A, B, D and E only
(2) A, C and E only
(3) A, D and E only
(4) A, B, C, D and E

125. Which of the following is a function of eosinophils?
(1) Phagocytosis
(2) Blood clotting
(3) Allergic reactions
(4) Osmotic balance

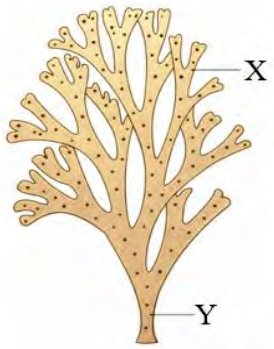
126. In an experiment to study DNA uptake in bacteria, which treatment increases competence?
(1) Treating with sodium chloride followed by heat shock.
(2) Treating with calcium ions followed by heat shock at 42°C.
(3) Treating with trichloroacetic acid.
(4) Treating with potassium ions at 0°C.

127. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The largely tropical Amazonian rain forest has the greatest biodiversity on earth.
Reason (R): Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable, which promotes niche specialisation and greater species diversity.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

[Contd...

128. The total number of cells in a mature embryo sac of a typical angiosperm is;
(1) 4.
(2) 8.
(3) 2.
(4) 7.

129. In the alga shown below;



(1) the structure 'X' is the air bladder that performs nitrogen fixation.
(2) the structure 'Y' is the holdfast that helps it to attach to the substratum.
(3) asexual reproduction occurs by pyriform non-motile zoospores.
(4) sexual reproduction occurs by pear-shaped gametes that possess flagella.

130. Given below are two statements:
Statement I: Auxins like IAA and indole butyric acid (IBA) have been isolated from plants, whereas NAA and 2,4-D are synthetic auxins.
Statement II: Auxins help to initiate rooting in stem cuttings, promote flowering in pineapples, and prevent fruit and leaf drop at early stages.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

131. In the human endocrine system, which gland is called a "lobular structure" located between lungs behind the sternum?
(1) Thyroid
(2) Adrenal
(3) Thymus
(4) Pineal

132. Read the following statements regarding the human respiratory system:
A. The conducting part clears inhaled air from foreign particles and humidifies it.
B. The exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.
C. Expiration occurs when intra-pulmonary pressure is lower than atmospheric pressure.
D. A healthy human breathes on average 12-16 times per minute.
E. The solubility of O_2 is 20-25 times higher than that of CO_2.
Which of the above statements is/are **correct**?
(1) A, B and C
(2) C and E only
(3) A, B, D and E
(4) A, B and D only

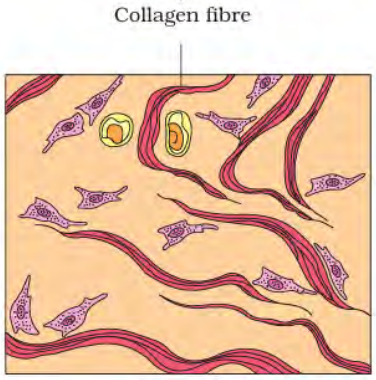
133. In a chemical analysis of living tissue, what does the "acid-soluble pool" primarily represent?
(1) Macromolecules such as proteins and polysaccharides
(2) Cytoplasmic composition with small molecular weight organic compounds
(3) Only inorganic elements like calcium and magnesium
(4) Membrane fragments in the form of vesicles

134. The organism that causes white spots on mustard leaves;
A. belongs to class ascomycetes.
B. has an aseptate and coenocytic mycelium.
C. is unicellular.
D. reproduces sexually by producing basidiospores.
E. cannot reproduce asexually.
How many of the above statements are **incorrect**?
(1) One only
(2) Two only
(3) Three only
(4) Four

135. Identify the **incorrect** statement about pea.
(1) Member of family Cruciferae.
(2) Aestivation in corolla is papilionaceous.
(3) Flower is zygomorphic.
(4) Leaf is modified into tendrils.

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136. Which of the following **correctly** identify the location of type of connective tissue shown in given below diagram?



(1) Bone
(2) Cartilage
(3) Dermis of Skin
(4) Cardiac muscles

137. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In a dihybrid cross, the F_2 phenotypic ratio of 9:3:3:1 indicates that the alleles of two genes are assorting independently.

Reason (R): The Law of Segregation states that when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

138. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Order)**	
(A)	*Petunia*	(I)	Poales
(B)	*Triticum*	(II)	Carnivora
(C)	*Panthera*	(III)	Polymoniales
(D)	Gorilla	(IV)	Primata

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-III, C-IV, D-II
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-III, B-I, C-II, D-IV

139. The spent dung slurry from a biogas plant;

(1) can be used as a fertiliser.
(2) is of no use.
(3) can be used as fuel.
(4) is supplied to nearby houses using pipelines.

140. Consider the following four statements (A-D) and select the **correct** option stating which ones are true (T) and which ones are false (F).

A. The portion of myofibril between two successive Z lines is a sarcomere.
B. The H-zone is the central part of thick filament not overlapped by thin filaments.
C. Red muscle fibres contain high myoglobin and few mitochondria.
D. White muscle fibres possess an extensive sarcoplasmic reticulum

	A	**B**	**C**	**D**
(1)	T	T	F	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	T	F	F	F

141. Identify the **incorrectly** matched pair.

(1) Penicillin - Was used to treat American soldiers wounded in World War II
(2) Wine - Produced by distillation of the fermented broth
(3) Biogas - Produced in STP during consumption of cellulosic material by methanogens
(4) Ethanol - Produced commercially in large fermentors

142. Which of the following statement is **incorrect** regarding cockroach morphology?

(1) Males bear a pair of short anal styles.
(2) Anal cerci are present in both sexes.
(3) The head is formed by the fusion of three segments.
(4) Exoskeleton has hardened plates called sclerites.

143. The condition where male and female flowers are present on different plants, thus preventing both autogamy and geitonogamy, is called:

(1) Monoecy.
(2) Dioecy.
(3) Cleistogamy.
(4) Chasmogamy.

144. If 100000 J of sunlight is incident on producers of a grassland ecosystem, how much energy is expected to reach upto the secondary consumers?

(1) 1000 J (2) 100 J
(3) 10 J (4) 1 J

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145. Given below are two statements:

Statement I: The C_3 plants respond to higher temperatures and show higher rate of photosynthesis while C_4 plants have a much lower temperature optimum.

Statement II: Tropical plants have a higher temperature optimum than the plants adapted to temperate climates.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Which of the following is **not** true regarding the human heart?

(1) The heart is a mesodermally derived organ.
(2) The SA node is called the pacemaker.
(3) The left atrium receives deoxygenated blood from the lungs.
(4) It is protected by a double walled membranous bag, pericardium, enclosing the pericardial fluid.

147. In a neuron, Nissl's granules are absent in the:

(1) Cell body (2) Dendrites
(3) Axon (4) Both (1) and (2)

148. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Human kidneys can produce urine nearly four times more concentrated than the initial filtrate formed.

Reason R: The counter current mechanism in Henle's loop and *vasa recta* helps maintain an increasing osmolarity towards the inner medullary interstitium.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

149. Identify the **incorrect** statement about the enzyme DNA ligase.

(1) The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase.
(2) It plays an important role in synthesis of daughter strand during DNA replication on the template strand with polarity $3' \rightarrow 5'$.
(3) It is an important tool in genetic engineering and recombinant DNA technology.
(4) When DNA is cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky-ends' and, these can be joined together (end-to-end) using DNA ligases.

150. In a standard 28-day menstrual cycle, If the "LH Surge" is artificially inhibited on Day 13, what is the most likely prediction for the events on Day 15-25?

(1) Immediate menstruation will occur on Day 15.
(2) The Graafian follicle will transform into the Corpus Luteum despite no ovulation.
(3) Progesterone levels will fail to rise, and the endometrium will not enter the secretory phase.
(4) Estrogen levels will remain at their peak indefinitely.

151. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The biosynthesis phase of photosynthesis is indirectly dependent on the photochemical phase.

Reason (R): The biosynthesis phase of photosynthesis requires ATP and $NADP^+$ produced during photochemical phase.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

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152. Which statement regarding enzymes is **correct**?
A. Enzymes function optimally within a narrow range of temperature and pH.
B. Competitive inhibitors often resemble the substrate in structure.
C. Raising substrate concentration infinitely will always increase reaction velocity indefinitely.
D. V_{max} reflects saturation of all available enzyme active sites.
(1) A and B only (2) B and C only
(3) A, B and D only (4) A, B, C and D

153. Bacteria that live in extreme salty areas are called:
(1) true bacteria.
(2) halophiles.
(3) thermoacidophiles.
(4) cyanobacteria.

154. A human population of 200 individuals in a region increased to 260 and then it reached an asymptote. If the growth rate is 0.5, the intrinsic rate of natural increase for this population is approximately;
(1) 0.55 (2) 0.01
(3) 0.32 (4) 1.5

155. Identify the correct statements regarding the double helix structure of DNA.
A. Each chain has one free end with a phosphate group, at the 3' C.
B. The bases in two strands are paired through phosphodiester bonds.
C. A purine always comes opposite to a pyrimidine.
D. The two chains are always coiled in a left-handed fashion.
E. The pitch of the helix is 3.4 nm and there are roughly 10 bp in each turn.
Choose the **correct** option.
(1) C, D and E
(2) C and E only
(3) A, B and C
(4) A and C only

156. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hydrolases	(I)	Joining of two molecules
(B)	Ligases	(II)	Cleavage of bonds by adding water
(C)	Transferases	(III)	Catalysis of oxidation-reduction
(D)	Oxidoreductases	(IV)	Transfer of a group (other than H)

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-II, C-I, D-III

157. Match **List-I** with **List-II.**

List-I (Animal)		List-II (Distinguishing feature)	
(A)	*Chameleon*	(I)	Oviparous with indirect development
(B)	*Salamandra*	(II)	Oviparous with direct development
(C)	*Neophron*	(III)	Presence of milk producing glands
(D)	*Macropus*	(IV)	Air sacs connected to lungs

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

158. A person is bitten by a venomous snake. The doctor immediately injects pre-formed antibodies against the toxin. This is an example of:
(1) Active immunity
(2) Natural immunity
(3) Passive immunity
(4) Innate immunity

159. Which of the following is **not** a part of brainstem?
(1) Midbrain
(2) Pons
(3) Medulla oblongata
(4) Cerebrum

160. Identify the **correct** pairing for the common Indian frog:
(1) Heart — Three-chambered (two auricles and one ventricle)
(2) Kidneys — Located anteriorly in the body cavity
(3) Skin — Rough and dry on the dorsal side
(4) Eyes — Protected by tympanum while in water

161. When $FADH_2$ donates the electrons to complex II, which complexes of ETS facilitate proton movement from mitochondrial matrix to the intermembrane space?
(1) I, III and IV (2) I, II and III
(3) III and IV only (4) I, II, III and IV

162. Which of the following statements about lymphoid organs is **correct**?
(1) Bone marrow and thymus are secondary lymphoid organs.
(2) The spleen acts as a filter for lymph.
(3) MALT constitutes about 50% of the lymphoid tissue in the human body.
(4) Antigens interact with lymphocytes primarily in primary lymphoid organs.

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163. Identify the **correct** statement about human blood grouping:
(1) A person with 'O' blood group is a universal recipient.
(2) AB blood group individuals have no antigens on RBCs.
(3) *Erythroblastosis foetalis* occurs in the first pregnancy due to Rh incompatibility.
(4) 'O' blood group can be donated to any other group.

164. Given below are two statements:
Statement I: In prokaryotes, the DNA is not scattered throughout the cell but is held with some positively charged proteins in a region termed as 'nucleoid'.
Statement II: In eukaryotes, the negatively charged DNA is wrapped around a positively charged histone octamer to form a structure called euchromatin.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

165. In an ecosystem comprising of a number of insects feeding on a single big tree, the pyramid of energy will be;
(1) inverted.
(2) spindle-shaped.
(3) upright.
(4) urn-shaped.

166. Identify the **correctly** matched pair regarding contraceptive methods:
(1) Diaphragm — Single-use disposable barrier
(2) Cervical cap — Reusable barrier often used with spermicides
(3) Female condom — Inserted by a trained nurse only
(4) Nirodh — Brand name of a copper IUD

167. Identify the **incorrect** statement about plant tissue culture:
(1) Micropropagation produces genetically identical plants called somaclones.
(2) Virus-free plants can be obtained by meristem culture.
(3) All plant tissues, including meristems, always harbour virus particles in infected plants.
(4) Protoplast fusion generate somatic hybrids.

168. Identify the **correctly** matched pair.
(1) *Parthenium* - Causes pollen allergy
(2) *Lupinus arcticus* - Dormant seed discovered near King Herod's palace
(3) Rice - Many ovules in an ovary
(4) Cashew - Produces true fruit

169. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In skeletal muscle contraction, the I-bands get reduced while A-bands retain their length.
Reason R: Contraction takes place by the sliding of thin filaments over thick filaments.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

170. Match **List-I** with **List-II.**

List-I (Treatment method)		**List-II (Mechanism Description)**	
(A)	Radiotherapy	(I)	Use of biological response modifiers like alpha-interferon
(B)	Chemotherapy	(II)	Tumor cells are lethally irradiated while protecting normal tissue
(C)	Immunotherapy	(III)	Surgical removal of localized cancerous mass
(D)	Surgery	(IV)	Use of drugs to kill cancer cells; often has side effects like hair loss

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-III, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-I, D-II

171. Due to which event of cell cycle, the parent cell temporarily possesses two sets of chromosomes?
(1) DNA replication in S phase
(2) Splitting of centromere in anaphase
(3) Disintegration of nuclear envelope in prophase
(4) Recombination between homologous chromosomes in pachytene of meiotic prophase I

[Contd...

172. Match **List-I** with **List-II** and select the **correct** combination.

List-I		List-II	
(A)	Squamous epithelium	(I)	Connect muscle to bones
(B)	Tendon	(II)	Connect bone to bone
(C)	Ligament	(III)	Present in walls of blood vessels
(D)	Cuboidal epithelium	(IV)	Present in ducts of glands

(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

173. Who proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances?
(1) Melvin Calvin
(2) J. A Bassham
(3) Cornelius Van Niel
(4) Jan Ingenhousz

174. The chief 'producers' in the oceans;
(1) lack cell wall.
(2) have a spore like stage in life cycle.
(3) are responsible for the formation of diatomaceous earth.
(4) are considered to be primitive relatives of animals.

175. Steroidal hormones in animal cells are synthesised by;
(1) Smooth Endoplasmic Reticulum.
(2) Lysosome.
(3) Mitochondria.
(4) Golgi complex.

176. Arrange the sub-stages of Prophase I of meiosis in the correct sequence.
A. Diplotene
B. Pachytene
C. Leptotene
D. Diakinesis
E. Zygotene
Choose the **correct** option.
(1) C → E → B → A → D
(2) C → B → E → A → D
(3) E → C → B → D → A
(4) C → E → A → B → D

177. During the light reaction of photosynthesis, protons accumulate in;
(1) chloroplast stroma.
(2) thylakoid lumen.
(3) intermembrane space of the chloroplast.
(4) perinuclear space of the nuclear membrane.

178. Arrange the following events of life cycle in pteridophytes in the correct sequence.
A. Formation of thalloid gametophyte
B. Meiosis in sporangial cells
C. Production of gametes
D. Production of spores
E. Formation of zygote
Choose the *most appropriate* answer from the options given below.
(1) C → E → A → B → D
(2) E → C → A → B → D
(3) C → A → E → B → D
(4) B → D → A → C → E

179. Identify the substrate level phosphorylation steps of aerobic respiration starting from the entry of glucose in it.
A. 3-phosphoglyceric acid → 2-phosphogycerate
B. Pyruvate → Acetyl CoA
C. Phosphoenolpyruvate → Pyruvate
D. Succinyl-CoA → Succinic acid
E. Fructose-6-phosphate → Fructose1, 6-bisphosphate
Choose the **correct** option.
(1) C and D
(2) B and C
(3) D and E
(4) C and E

180. Which of the following is **not** a benefit of transgenic animals in biological research?
(1) Ensuring that the animal never develops a disease
(2) Study of normal physiology and development
(3) Producing biological products like alpha-1-antitrypsin
(4) Testing the safety of vaccines

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